ZEIT4230: Prototype Design 2

System Description and Requirements

The system is a wearable bone-conduction digital hearing aid designed to improve speech intelligibility, listening comfort, and situational awareness for users with mild to moderate hearing loss[1]. The device is worn on the side of the head and transmits processed audio signals to the cochlea via bone conduction, allowing sound perception.

The system captures audio from a dual-microphone array, enabling spatial audio acquisition and improved noise discrimination. These signals are converted to digital form and processed using real-time Digital Signal Processing (DSP) including gain control, noise reduction, and feedback suppression, to enhance speech clarity while maintaining safe and comfortable hearing levels. The processed signal is then output through a bone-conduction transducer.

The system supports multiple audio input modalities to address diverse listening environments. In addition to ambient microphone input, it includes:

- Bluetooth Low Energy (BLE) audio streaming, enabling direct wireless audio reception from external devices such as smartphones, and
- Telecoil (T-coil) input, allowing reception of electromagnetic signals from hearing loop systems commonly deployed in public venues.

These features ensure that the system can maintain high speech intelligibility in challenging acoustic environments.

Hearing loss[1] – Loss of hearing including sensorineural, conductive, or mixed.

Mode	**Description**	**Entry Condition**	**Exit Condition**	**Key Behaviour**
OFF (M0)	No system power	User holds power button ≥ 1 s	Power button pressed	No operation
STARTUP (M1)	System initialisation	Transition from OFF to ON	Initialisation complete (<1 s)	Hardware + DSP initialised
NORMAL (M2)	Active hearing assistance	Device worn and system ON	Device removed or fault	Audio processing active
SLEEP (M3)	Low-power standby	No proximity detected for ≥ 5 s	Proximity detected	Minimal power consumption
FAULT (M4)	Unsafe or invalid condition	Fault detected	Reset or recovery	Audio muted, user alerted

Use Case ID:	***Use Case Name:***	***Level:***
UC - 01	Digital Hearing Aid: Normal Operation	System

Goal:	Enable continuous, real-time enhancement of speech and environmental audio with safe output levels across varying acoustic environments.
Primary Actor:	End User

System in Use Story:

When worn, the system detects sustained skin proximity and transitions from SLEEP to STARTUP to NORMAL within 1 second. The dual-microphone array captures ambient sound, which is digitised and processed in real time using DSP to enhance speech intelligibility and suppress noise.

The user may adjust volume or switch between Microphone (M), Bluetooth (B), and Telecoil (T) modes. The system prioritises audio sources based on the selected mode.

When the device is removed, the system detects loss of proximity and transitions to SLEEP mode within 5 seconds to conserve power.

Basic Flow of Events:

1. The user places the hearing aid on the side of their affected ear.
2. The proximity sensor detects skin contact sustained for ≥ 500 ms
3. The system transitions from SLEEP to STARTUP to NORMAL.
4. Microphones capture audio (20 Hz – 8kHz).
5. Audio is digitised (16-bit, 48kHz sample rate).
6. DSP performs:
    - Noise reduction (≥ 10 dB SNR)
    - Automatic gain control
    - Feedback suppression
7. Processed signal sent to bone-conduction transducer
8. User adjusts volume via rotary encoder (response time ≤ 100 ms)
9. System maintains continuous operation with latency ≤ 50 ms
10. Device removed with no proximity detected for ≥ 5 s
11. System transitions to SLEEP.

Flow Extensions (Mode-Based Behaviour)

- In Bluetooth mode
    1. Activated via user input.
    2. System disables microphone and telecoil input.
    3. Streams BLE audio at ≥ 128 kbps.
    4. Maintains latency ≤ 100 ms.
- In Telecoil mode
    1. Activated via user input.
    2. System disables microphone and Bluetooth input.
    3. Receives inductive signal (300 Hz – 5kHz).
    4. Converts to audio output.

Alternative Flows (Fault and edge cases)

- Bluetooth connection loss

- Detection time ≤ 2 s.
- System automatically switches to microphone mode within ≤ 500 ms.

- False proximity trigger
  - Requires ≥500 ms sustained detection before state transition.
  - Prevents rapid oscillation between SLEEP and NORMAL.
- Audio overload / unsafe output
  - output exceeds safe threshold (85 dB SPL)
  - System limits gain within ≤ 50 ms.
- Software (DSP) fault
  - System transitions to FAULT.
  - Audio output is muted.

Use Case ID:	***Use Case Name:***	***Level:***
UC - 02	Power Charging and Balancing	System
Goal:	Enable safe charging, monitoring, and balancing of the internal 2s1p 26650 $LiFePO_4$ battery pack while preventing unsafe operating conditions and ensuring reliable system behaviour without dedicated visual indicators.	
Primary Actor:	End User	

System in Use Story:

During normal operation, the system continuously monitors the voltage and operating condition of the internal 2s1p 26650 $LiFePO_4$ battery pack (3.2 V nominal per cell, 3 Ah capacity) at a sampling rate of ≥ 1 Hz. When the battery state-of-charge falls below defined thresholds, the system alerts the user via audio output behaviour and initiates a controlled shutdown at the critical threshold to prevent deep discharge.

To recharge, the user connects the device to an external power source via the designated charging interface. Charging is only permitted when the system is in SLEEP mode (no proximity detected for ≥ 5 s).

Upon detection of a valid external power source while in SLEEP mode, the system begins charging within ≤ 500 ms. The system applies a $LiFePO_4$ charging profile (constant current followed by constant voltage) while continuously monitoring individual cell voltages.

To ensure safe operation and battery longevity, the system performs cell balancing when the voltage difference between cells exceeds 100 mV. Once charging is complete, the system safely terminates charging and returns to SLEEP mode, remaining there until proximity is detected and normal operation resumes.

Basic Flow of Events:

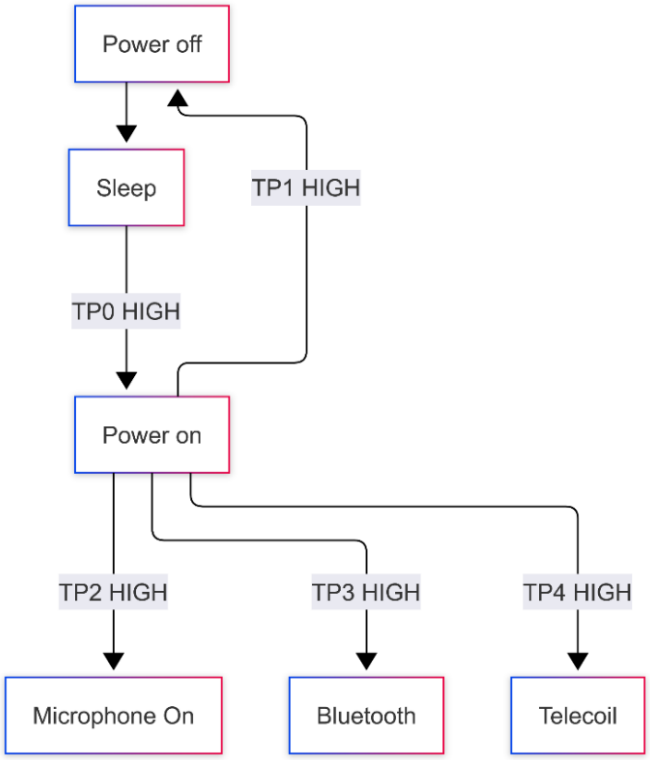
1. System monitors battery voltage continuously (≥1 Hz sampling)
2. Battery threshold detected:
   - Low (<20%) - user alerted via audio attenuation or tone
   - Critical (<10%) - system prepares shutdown
3. At critical threshold, system performs controlled shutdown and enters OFF mode

4. User connects external power source
5. System detects valid input power
6. System enters SLEEP mode if no proximity is detected for ≥ 5 s
7. System begins charging within 500ms
8. System monitors individual cell voltages and balances voltage differences >100 mV
9. Charging continues until termination current threshold reached
10. System remains in SLEEP until user prompts NORMAL usage

Flow Extensions (Conditional branches relating to a variable state within the case)

- Cell overvoltage (> 3.65V)
    1. Charging terminated (≤ 50 ms)
    2. System enters FAULT mode, audio disabled.
- Cell undervoltage (< 2.5V)
    1. System enters FAULT mode, prevents further discharge.
- Overcurrent
    1. Charging/discharge halted (≤50 ms)
    2. System enters FAULT mode
- Thermal fault (> 60°C)
    1. Charging stopped immediately
    2. System enters FAULT mode
- Invalid charger input
    1. Charging not initiated
    2. System remains in SLEEP or OFF mode

System Operating States and Responses to Expected and Unexpected Inputs



System States	Subsystem							
	Power Supply	User Control	System Monitoring	DSP	Audio Input	Bluetooth	Telecoil	Audio Output
Power Off								
Sleep								
Power On								
Microphone On								
Bluetooth								
Telecoil								

Note:

- **Green indicates subsystems in the given state.**
- **Orange indicates disabled or inactive subsystems.**

System-level Requirements

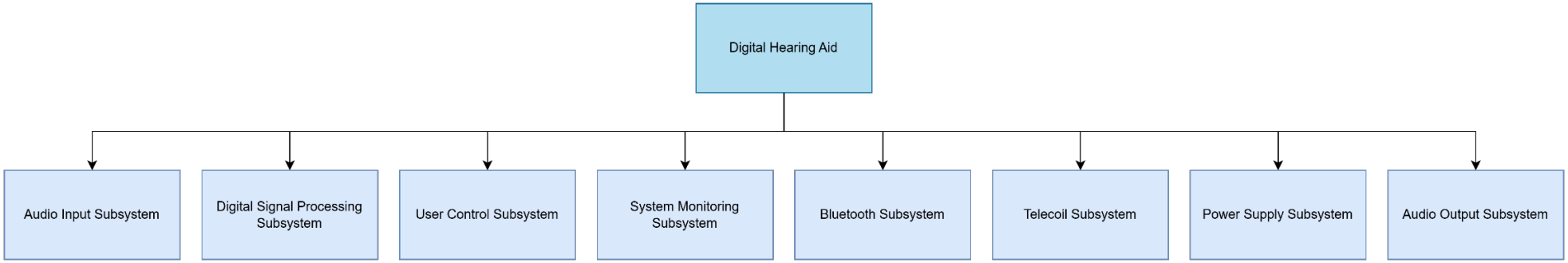
ID	Requirement Statement	Requirement Category	Verification	Justification / Design Decision	Input to the System	Output from the System
Functional Requirements						
FUN-01	The system shall amplify incoming audio signals across the 20 Hz – 20 kHz range.	Audio	Frequency response analysis.	Covers critical speech frequencies (human hearing range).	Acoustic audio input	Amplified audio signal
FUN-02	The system shall dynamically adjust gain based on input sound level.	Audio	Variable sound pressure level (SPL) test	Prevents discomfort to the user and distortion of the audio signal.	Varying audio signal levels.	Gain-adjusted audio
FUN-03	The system shall reduce background noise using digital signal processing.	Audio	Signal-to-noise ratio (SNR) measurement	Improves speech clarity.	Noisy audio input	Filtered audio signal
FUN-04	The system shall detect and suppress acoustic feedback.	Audio	Feedback loop test	Prevents whistling artifacts	Feedback-contaminated audio	Stable audio output
FUN-05	The system shall support Bluetooth Low Energy (BLE) streaming.	Bluetooth	Bluetooth pairing and streaming test.	Enables smartphone integration.	Bluetooth audio stream.	Decoded audio signal
FUN-06	The system shall support telecoil input for hearing loop systems.	Telecoil	Inductive coupling test	Enables use in assisted listening environments	Magnetic field (telecoil input)	Converted electrical audio
FUN-07	The system shall include at least one microphone	Audio	Functional test	Required for environmental awareness	Acoustic sound waves	Electrical audio signal

	for ambient sound capture.					
FUN-08	The system shall output processed audio to a speaker/receiver.	Output Audio	Audio output validation	Core function of a hearing aid	Processed audio signal	Acoustic sound output
Performance Requirements						
PER-01	End-to-end audio latency shall be < 50 ms.	Audio	Latency measurement	Prevents perceptual delay	Audio input signal	Real-time audio output
PER-02	Total harmonic distortion (THD) shall be < 3% at nominal levels.	Output	Audio analysis	Ensures sound quality	Audio signal	Low-distortion output
PER-03	The system shall support dual-microphone input for spatial audio capture.	Audio	Functional test	Enables beamforming and noise reduction	Dual mic signals	Multi-channel audio
Interface Requirements						
INT-01	The system shall allow the user to control volume and modes (Bluetooth, telecoil, or ambient audio).	User Interface	UI testing	Ensures usability	User input (buttons/app)	Selected audio stream and or Amplitude (gain) modified audio signal
INT-02	The system shall detect whether the device is worn (proximity detection).	System Control	Sensor validation	Enables power saving and automation	Proximity/skin signal	Wear-state flag
INT-03	The system shall support audio sources (mic, Bluetooth, telecoil)	Connectivity	Source switching test	Seamless user experience	Multiple audio inputs	Selected audio stream

	independently with prioritisation.					
Mechanical Requirements						
MEC-01	The earpiece shall fit within standard hearing aid dimensions (< 60 mm).	Mechanical	Physical measurement	Ensures comfort and wearability	N/A	Compact physical device
MEC-02	The head band shall form to the shape of the wearer's head, just like bone conductor earphones.	Mechanical	Physical measurement of whether the headband digs in or cannot be heard	Comfort and usability	User's head shape	Ergonomic Headpiece that fits head shape
MEC-02	The headband shall be robust and stay on the user's head during general day to day activities, including fitness	Mechanical	Wear the hearing aid for a day and record any observations.	Ensures the device	Physical activity and movement	Headpiece stays in the right position on the user's head
Power Requirements						
POW-01	The system shall operate continuously for ≥ 24 hours on a single charge.	Power	Battery discharge test	Meets daily usage expectations	Battery energy	Sustained operation
POW-02	The system shall enter low-power mode when not worn.	System Control	Power consumption test	Extends battery life	Wear-state signal	Reduced power consumption
Environmental Requirements						

ENV-01	The system shall operate between 0°C – 40°C and resist humidity.	Environmental	Environmental testing	Ensures real-world reliability	Environmental conditions	Normal operation
Safety Requirements						
SAF-01	The system shall comply with IEC 60118-13 device safety standards.	Safety	Compliance testing	Regulatory requirement	N/A	Certified safe operation
SAF-02	The system shall limit the maximum output sound pressure level to safe limits.	Output	SPL measurement	Prevents hearing damage	Audio signal	Safe-level audio output

Subsystem Breakdown



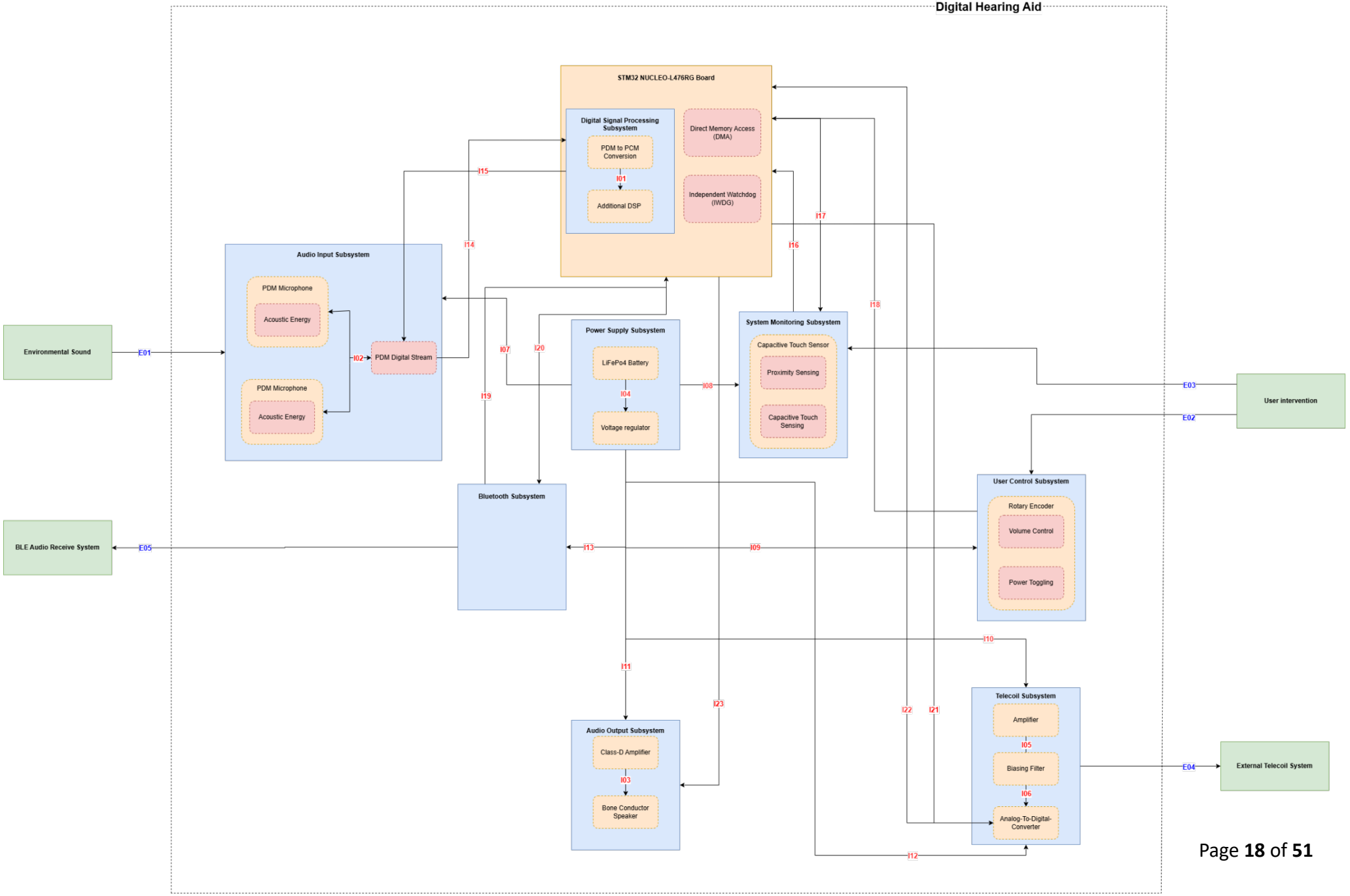
Subsystem	Responsible member
Audio input	Mariya
Digital signal processing	Mariya
User Control	Anna
System Monitoring	Anna
Bluetooth	Ryan
Telecoil	Ryan
Audio Output	Naomi
Power Supply	All

Pin Name	Type	Direction	Description	Bus	ID
DFSDM1_CKOUT	Digital	STM32 - Microphone	Serial Clock line	DFSDM	I15
DFSDM1_DATIN3	Digital	Microphone - STM32	Data line		I14
3V3	Common	Power - Microphone	Power line		I07
Ground	Common	Power - Microphone	Power line		
I2C1_SCL	Digital	STM32 - Capacitive touch	Serial Clock line	I2C	I17
I2C1_SDA	Digital	Capacitive touch - STM32	Data line		I16
3V3	Common	Power – Capacitive touch	Power line		I08
Ground	Common	Power – Capacitive touch	Power line		
TIM2_CH1	Digital	Rotary encoder - STM32	Data line	GPIO	I18
TIM2_CH2	Digital	Rotary encoder - STM32	Data line		
GPIO_IN	Digital	Rotary encoder - STM32	Push switch		
5V	Common	Power - STM32	Power line		I09
Ground	Common	Power - STM32	Power line		
SAI1_SD_A	Digital	Bluetooth - STM32	Data line	SAI (I2S)	I19
SAI1_SCK_A	Digital	STM32 - Bluetooth	Serial Clock line		I20
SAI1_FS_A	Digital	STM32 - Bluetooth	Left/Right Channel Clock line		

5V	Common	Power - Bluetooth	Power line		I13
Ground	Common	Power - Bluetooth	Power line		
Processed RF signal	Analog	Telecoil – Analog frontend	Analog signal received by the Telecoil	Analog	I05
5V	Common	Power – Analog frontend	Power line		I12
Ground	Common	Power – Analog frontend	Power line		
Analog_in (R_in and L_in shorted)	Analog	Analog frontend - ADC	Processed analog signal in	Analog	I06
SAI2_SD_A	Digital	ADC - STM32	Data line	SAI (I2S)	I22
SAI2_SCK_A	Digital	STM32 - ADC	Serial Clock line		I21
SAI2_FS_A	Digital	STM32 - ADC	Left/Right Channel Clock line		
3V3	Common	Power - ADC	Digital Power line		I10
5V	Common	Power - ADC	Analog Power line		
Digital Ground	Common	Power - ADC	Digital Power line		
Analog Ground	Common	Power - ADC	Analog Power line		
SAI2_SD_B	Digital	STM32 – I2S amplifier	Data line	SAI (I2S)	I23
SAI2_SCK_B	Digital	STM32 – I2S amplifier	Serial Clock line		
SAI2_FS_B	Digital	STM32 – I2S Amplifier	Left/Right Channel Clock line		
5V	Common	Power – I2S amplifier	Power line		I11
Ground	Common	Power – I2S amplifier	Power line		
Analog_OUT	Analog	I2S amplifier – Bone conductor	Analog signal out	Analog	I03

ID	Description
E01	Environmental acoustic sound waves within human hearing range (20Hz-20kHz) gets captured by the MEMS microphones.

E02	User interacts via physical controls (rotary encoder and its built-in push button) providing digital input signals to the system for volume adjustment and power control.
E03	User interacts via system monitoring controls (capacitive touch sensor) causing a change in capacitance which triggers proximity sensing (detect whether hearing aid is in the ear) and capacitive touch sensing (to choose between 3 modes: normal hearing, Bluetooth and telecoil).
E04	External telecoil system provides an analog electromagnetic audio signal.
E05	Bluetooth Low Energy (BLE) audio receive system transmits digital audio and control data wirelessly to the system via BLE communication interface.


Digital Hearing Aid
STM32 NUCLEO-L476RG Board
Digital Signal Processing Subsystem
PDM to PCM Conversion
Additional DSP
Direct Memory Access (DMA)
Independent Watchdog (IWDG)
Audio Input Subsystem
PDM Microphone
Acoustic Energy
PDM Microphone
Acoustic Energy
PDM Digital Stream
Power Supply Subsystem
LiFePo4 Battery
Voltage regulator
System Monitoring Subsystem
Capacitive Touch Sensor
Proximity Sensing
Capacitive Touch Sensing
Environmental Sound
E01
User intervention
E03
E02
Bluetooth Subsystem
BLE Audio Receive System
E05
User Control Subsystem
Rotary Encoder
Volume Control
Power Toggling
Telecoil Subsystem
Amplifier
Biasing Filter
Analog-To-Digital-Converter
E04
External Telecoil System
Audio Output Subsystem
Class-D Amplifier
Bone Conductor Speaker
I01
I02
I03
I04
I05
I06
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I22
I23

Subsystem Requirements

Subsystem(s)	Audio input subsystem, Digital Signal Processing subsystem
Group Member	Mariya

ID	Requirement Statement	Verification	Justification/ Design Decision	Input	Output	System Requirement Satisfied
1 Audio Input Subsystem						
Functional Requirement						
1.1	Each microphone shall output a 1-bit PDM digital stream.	Measure output using logic analyser.	PDM output is native to MEMS microphone architecture (datasheet).	Acoustic signal	1-bit PDM stream	FUN-07
1.2	Shall acquire 2 microphone channels simultaneously with time aligned sampling.	Record both the channels and compare the time stamps.	Required to preserve time-of-arrival differences for beamforming.	Acoustic signal	Time-aligned PDM streams	PER-03
1.3	All microphones shall share a common clock source.	Verify shared clock using oscilloscope.	Common clock ensures synchronised sampling and phase coherence.	Clock signal	Synchronous PDM output	PER-03
1.4	During normal operation, system shall provide	Continuous recording test (check for	Required for uninterrupted real-time processing.	Acoustic signal	Continuous PDM stream	PER-01

	continuous PDM output.	missing samples).				
1.5	The subsystem shall provide independent data streams for each microphone channel.	Verify DFSDM channel mapping.	Independent channels required for multi-channel processing.	Acoustic signal	Two independent PDM streams	PER-03
Acoustic Performance Requirement						
1.6	The subsystem shall achieve a SNR $\geq 64dB$ per microphone (A-weighted).	Measure RMS signal versus noise in a quiet environment.	Datasheet SNR ensures sufficient signal quality.	Acoustic signal and noise.	Clean digital signal	FUN-01, FUN-03
1.7	The subsystem shall exhibit total harmonic distortion (THD) $\leq 1\%$ at 94dB SPL input.	Perform FFT analysis of 1kHz tone at 94dB SPL.	Datasheet THD ensures minimal distortion.	Pure tone input.	Low harmonic distortion	PER-02
1.8	The subsystem shall support an acoustic overload point (AOP) $\geq 122.5dB$ SPL without clipping.	Increase SPL until clipping is observed.	Datasheet AOP ensures handling of loud sounds.	High SPL sound.	Non-clipped signal	FUN-01, ENV-01
1.9	The system shall have a microphone sensitivity of $-26dBFS \pm 3dB$.	Measure output at known SPL (e.g. 94 dB).	Datasheet sensitivity used for calibration and gain design.	Known SPL input.	Expected dBFS output	FUN-01

Beamforming Requirement						
1.10	The subsystem shall arrange two microphones in a linear end-fire configuration.	Inspect PCB layout	End-fire array enables directional listening along axis.	Acoustic signal	Spatially separated signals	PER-03
1.11	The microphone shall be aligned along the forward-facing direction of the system to prioritise sound arriving at the front of the user.	Verify physical orientation relative to device front.	Hearing aid prioritises front-facing speech .	Acoustic signal	Directional sensitivity	FUN-03
1.12	Microphone spacing shall be between 10mm and 20mm and known within a tolerance of $\pm 0.5mm$.	Measure spacing via CAD or callipers.	Accurate spacing required for delay calculation in beamforming.	Acoustic signal	Known spatial separation	PER-03
1.13	Both microphone shall be driven by a common PDM clock source.	Verify shared clock via oscilloscope.	Required to maintain synchronisation between channels.	Clock signal	Synchronous signals	PER-03
1.14	The inter-channel timing mismatch shall be ≤ 1 PDM clock cycle.	Measure timing difference using logic analyser .	Ensures accurate time-delay estimation.	PDM signals	Aligned channels	PER-03

1.15	The frequency response between microphones shall be $\leq \pm 3dB$.	Compare frequency response of both microphones.	Matching prevents distortion in beam pattern.	Acoustic signal	Matched frequency response	PER-03
1.16	The sensitivity variation between microphones shall be $\leq \pm 3dB$.	Compare output levels under same input.	Matches datasheet tolerance and ensures balance.	Acoustic signal	Equal amplitude signals	PER-03
Electrical and Power Requirement						
1.17	Each microphone in the subsystem shall operate at a supply voltage of 1.8V (Typically).	Measure supply voltage using multimeter.	Matches microphone operating voltage.	Power supply	Stable operation	POW-01
1.18	Total subsystem power consumption shall not exceed 3mW.	Measure current and calculate power.	Derived from datasheet current consumption with margin.	Power supply	≤ 3 mW consumption	POW-01
Environmental Requirement						
1.19	The subsystem shall function in acoustic environments from 30dB SPL to 100dB SPL.	Test with controlled SPL audio source.	Covers typical human speech environments.	Acoustic signal (30–100 dB SPL).	Functional audio output	ENV-01
2 Digital Signal Processing Subsystem						
2.1	The subsystem shall convert incoming PDM signals to PCM audio using digital	Capture output PCM data and verify sampling rate and bit depth.	Required as PDM microphones output 1-bit data; DSP algorithms require PCM (from ST AN3998)	PDM bitstream	PCM audio signal	FUN-01, PER-01

	filtering and decimation.					
2.2	The subsystem shall implement automatic gain control (AGC) to amplify low-level audio signals.	Input low amplitude signal and measure output gain increase.	Required to improve audibility of quiet sounds in hearing aid use.	Low-level PCM signal	Amplified signal	FUN-02
2.3	The subsystem shall reduce background noise using digital filtering techniques.	Input noisy signal and compare SNR before and after processing.	Improves speech clarity by removing steady background noise.	Noisy PCM signal	Noise-reduced signal	FUN-03
2.4	The subsystem shall enhance speech signals within the 300 Hz–4 kHz frequency band.	Apply frequency analysis (FFT) before and after processing.	Improves clarity	PCM signal	Speech-enhanced signal	FUN-01, FUN-03
2.5	The subsystem shall implement delay-and-sum beamforming using two microphone channels.	Input directional sound and verify increased response in target direction.	Enables directional listening and noise suppression	Two-channel PCM input	Directionally enhanced signal	PER-03, FUN-03

Justification of Chosen Development Board

The Adafruit PDM MEMS Microphone Breakout development board was chosen since it integrates a digital MEMS microphone with a pulse density modulation (PDM) output interface. The board uses DFSDM peripheral to connect to the STM32 NUCLEO-L476RG board.

Subsystem Requirement	How the board satisfies it
1.1	Provides 1 bit output [22], directly compatible with STM32 DFSDM
1.5	Multiple breakout boards can be used for multi-channel acquisition.
1.3	Supports external clock input required for synchronised sampling.

As the board satisfies the subsystem requirements of Audio input as quoted in the above table, it eliminates the need for external ADC's which reduces system complexity and power consumption. If an analog microphone were to be used, the captured sound would have had to go through an amplifier then an ADC after only which it can digitally signal processed. With a PDM microphone the microphone already outputs a digital signal without requiring external ADC or analog front-end since it is handled by the DFSDM peripheral [8]. Also an external ADC would have consumed a lot more additional current.

The microphone supports shared clock operation and left/right channel multiplexing through data pin selection [23] as outlined in sub system requirements 1.2, 1.3, 1.13 and 1.14. This enables the two microphones to operate on a common clock and time aligned sampling required for beamforming. This also therefore supports the beamforming requirements as well as system requirement PER-03.

Since the breakout board integrates a digital MEMS microphone IC (MP34DT05) whose acoustic performance meets critical acoustic specifications (SNR, THD, AOP and sensitivity) [22] as outlined in subsystem requirements 1.6 through to 1.9, it ensures high speech clarity (FUN-01 and FUN-03), low distortion (PER-02) and adequate dynamic range for real world environments (ENV-01). The board also satisfies the low power requirements of the system where two microphone breakouts consume total power approximately equal to 2.4mW which satisfies the subsystem limit of less than 3mW [21](subsystem requirement 1.18) which supports 24 hour operation. The breakout board enables beamforming since it can be flexibly connected within allowable 10-20mm spacing.

Also, the breakout board allows for a good integration with the STM32 NUCLEO-L476RG low power board which has a DFSDM peripheral through which the microphone is connected. The hardware decimation reduces CPU load thus enabling requirements PER-01 [8].
In comparison with alternative digital microphones such as I2S microphones, the PDM microphone was chosen as there is other subsystems such as the Bluetooth, telecoil and output amplifier which rely on the I2S peripheral to connect the NUCLEO board [22]. Given the NUCLEO-L476RG has DFSDM peripheral, the PDM microphone was connected via the same to allow for other critical applications even though it could be connected via I2S and SAI peripheral.

The limitation of choosing the breakout board is, the two boards combined (14.0mm x 12.8mm x 2.8mm [21]) is larger than the final hearing aid requirement (MEC-01), hence can only be used for prototyping, with final design only integrating the bare MEMS IC. However it allows rapid system validation, with the intention of transitioning to a custom PCB in later design stages.

Subsystem Requirements

Subsystem(s)	Audio output subsystem
Group Member	Naomi

ID	Requirement Statement	Verification	Justification / Design Decision	Input	Output	Linked System Req
7.1	The subsystem shall receive and amplify the received signal.	Oscilloscope test, comparing electrical signal before and after passing through amplifier	Ensure sufficient audio amplitude for user to hear	Electrical audio signal	Amplified electrical audio signal	FUN-01,08
7.2	The subsystem shall amplify the electrical signal variably based on the circuit setup	Using a multimeter to verify the impedance and voltage input o the amplifier and oscilloscope for the amplitude of the output signal	Robust output repairability	Electrical audio signal	Amplified electrical audio signal	FUN-02
7.3	The subsystem shall output the audio from one of either the telecoil input, the Bluetooth input or the microphone input	Streaming Validation Test	Ensures the system fulfills all 3 modes exclusively	Electrical audio signal	Mechanical Audio signal	FUN-05,06,07 INT-03
7.4	The subsystem shall amplify the microphone input audio less than 50 ms after real time.	Oscilloscope test, probe at the microphone input and at the electrical output.	User experiences no lag in audio.	Microphone electrical audio signal	Amplified electrical audio signal	PER-01
7.5	The subsystem shall not output any noise whilst not engaged by the other subsystems	Oscilloscope test	Signal should have no underlying noise	The absence of Electrical audio signal	Amplified electrical audio signal	FUN-04

7.6	The subsystem shall have no more than 3% harmonic distortion at nominal levels.	oscilloscope + FFT test and analysis	Clear audio signal for user	Input audio electrical signal	Output electrical audio signal	PER-02
7.7	The subsystem housing shall be no greater than 60mm in any direction	Physical measurement	Ensures comfort and wearability	N/A	Compact physical device	MEC-01
7.8	The subsystem housing shall adapt to the shape of a variety of heads, tight enough that it stays on during physical activity but not too tight to cause headaches	Physical measurement and report observations across the span of a day of wear	Ensures the device is ergonomic and practical	N/A	Physical Housing comfort with bone conductors	MEC-02,03
7.9	The subsystem shall not exceed a level of vibration that discomforts the user	User hearing test report	Ensure the user is comfortable and no hearing damage is caused over long term use.	Output audio signal	Audio signal through skull	SAF-02

Component	**Function in System**	**Key Specs / Metrics**	**Subsystem**	**Addresses Requirements**	**Verification**	**Justification / Engineering Rationale**
MAX98357 I2S Amplifier Module	Amplify the electrical audio signal from the STM32	I^2S interface Output Power: 8O 1.8W (at 12 dB), Built-in DAC and amplifier in the breakout board, Variable gain	Output	FUN-01, 04, 08, PER-01, 02	oscilloscope, FFT, multimeter	Eliminates need for external DAC, low cost, improves simplicity, 8O 1.8 W is compatible with exciter speaker, class D amplifiers being more efficient than traditional amplifers is desirable for battery operating hearing aids, variable gain allows for further post-processing of the output signal before the signal gets converted to vibrations.
8 Ohms Exciter	Convert the electrical	Dimensions: 14mm x 21.5mm / 0.6" x .8"	Output	FUN-02, FUN-05,06,07	User observation,	1.5W sufficient to produce strong vibrations for bone conduction, maximum power

Speaker 1 W 300 Hz ~ 19 kHz	signal from the amplifier into vibrations heard through the user's skull	Weight: 9.6g Power handling: 1 watt RMS/2 watt max Le: 1.26 mH Impedance: 8 ohm Re: 5.8 ohms Frequency response: 300-19,000 Hz (but will vary with what you use as the transducer surface) Fs: 1,600 Hz SPL: 90.1 dB 1W/1m (but will vary with what you use as the transducer surface)		INT-03, SAF-02	sound level meter	receivable is 2W which is perfect for the 1.8 W amplifier module, human hearing range is 20 – 20k Hz, bone conductor includes 300 – 19k Hz which is 93.5% of the typical human hearing range and covers most of the human speech range. It is at risk of clipping in the lower range (<300Hz) unless some audio processing is done to mitigate this clipping.
3D printed bone conductor headband	Housing for the components against the user's head	PLA filament 139 x 110 x 48 mm (creative outlier free dimensions)	Output	MEC-01, 02, 03	User observation	PLA filament (returns to shape well, but sturdier than TPU and PETG), less pedantic than other filaments for printing. I plan on 3D scanning a classmate's bone conductors, refining the shape in Blender, including attachments for the electronic components then 3D printing the model.

Subsystem(s)	Bluetooth subsystem, Telecoil subsystem
Group Member	Ryan

ID	Requirement Statement	Verification	Justification / Design Decision	Input	Output	Linked System Req
1.1	The subsystem shall initialise Bluetooth and telecoil interfaces within 3 seconds of power-on in M1.	Startup timing test	Ensures responsive system startup	Power-on event	Initialised interfaces	FUN-05, FUN-06
1.2	The subsystem shall establish a Bluetooth audio connection within 5 seconds of entering M1 when a paired device is available.	Connection timing test	User experience requirement	BLE pairing request	Active BLE link	FUN-05
1.3	The subsystem shall continuously monitor for telecoil signal presence in M2 and M3.	Signal detection test	Enables automatic accessibility	Magnetic field input	Detection flag	FUN-06
1.4	The subsystem shall receive and decode Bluetooth audio streams in M2.	Streaming validation	Core wireless audio function	BLE audio stream	PCM audio data	FUN-05, FUN-08
1.5	The subsystem shall acquire and condition telecoil signals into a usable audio signal in M2.	Signal chain test	Enables hearing loop functionality	Magnetic input	Conditioned audio signal	FUN-06
1.6	The subsystem shall provide both Bluetooth and telecoil audio streams to the main processing unit via I^2S.	Interface test (logic analyser)	Standardised digital interface	Audio streams	I^2S audio data	INT-03
1.7	The subsystem shall arbitrate between Bluetooth and telecoil inputs using priority logic (Bluetooth > telecoil).	Arbitration test	Prevents conflicting audio	Multiple inputs	Selected stream	INT-03
1.8	The subsystem shall switch between active audio sources within 100 ms of a priority change.	Timing test	Ensures seamless user experience	Input change event	Updated audio output	PER-01
1.9	Bluetooth audio latency shall not exceed 100 ms from reception to I^2S output.	Latency measurement	Maintains usability	BLE stream	Timely PCM output	PER-01

1.10	Telecoil signal path shall achieve ≥ 20 dB SNR under nominal field strength conditions.	SNR measurement	Ensures intelligibility	Magnetic input	Clean audio	FUN-03
1.11	The subsystem shall support audio sampling rates of 16 kHz and 48 kHz.	Sampling validation	Compatibility with system DSP	Audio stream	PCM data	FUN-01
1.12	Idle mode power consumption shall be ≤ 10 mW.	Power measurement	Battery life optimisation	Idle state	Reduced consumption	POW-01
1.13	The subsystem shall interface with the main processor using I^2S (BCLK, LRCLK, SD).	Signal validation	Industry-standard audio interface	Digital clock + data	PCM stream	INT-03
1.14	The subsystem shall receive control signals for mode selection and enable/disable commands.	Control test	Enables user/system control	GPIO/UART commands	Mode changes	INT-01
1.15	The subsystem shall output a status signal indicating active source (Bluetooth or telecoil).	Status validation	Debugging and system awareness	Internal state	Status flag	INT-02
1.16	The subsystem shall enter Fault Mode (M4) within 100 ms of detecting Bluetooth link loss or invalid telecoil signal.	Fault injection test	Ensures rapid fault detection	Link loss / invalid signal	Fault state flag	FUN-05, FUN-06
1.17	The subsystem shall mute its audio output within 50 ms upon entering M4.	Timing test	Prevents audible artifacts	Fault condition	Muted output	SAF-02
1.18	The subsystem shall attempt reconnection to a previously paired Bluetooth device every 2 seconds in M4.	Retry timing test	Improves robustness	Fault state	Reconnection attempts	FUN-05
1.19	The subsystem shall automatically exit M4 and return to M2 upon detection of a valid audio source within 2 seconds.	Recovery test	Enables seamless recovery	Valid signal	Restored operation	INT-03
1.20	The subsystem shall transition to Idle Mode (M3) if no valid input is detected for 5 seconds in M4.	Timeout test	Prevents unnecessary power drain	No signal	Idle state	POW-01

Justification of Chosen Development Board

Component	Function in System	Key Specs / Metrics	Subsystem	Addresses Requirements	Verification	Justification / Engineering Rationale
BT401v3 Bluetooth Module	Wireless audio interface (BLE audio in/out)	BLE audio support, I²S interface, low power (~tens of mW), integrated stack	Connectivity	FUN-05, INT-03, PER-01, 1.2, 1.4, 1.9	BLE streaming + latency test	Integrated BLE + audio stack reduces firmware complexity and risk. Native I²S aligns directly with STM32 SAI, enabling deterministic audio transfer. Chosen over discrete BLE SoCs due to faster development and lower integration risk.
Knowles 5100-253447 Telecoil	Magnetic field sensing (hearing loop input)	High sensitivity inductive pickup, small form factor	Connectivity (Telecoil)	FUN-06, FUN-03, 1.3, 1.5, 1.10	Inductive coupling + SNR test	Industry-standard telecoil with proven performance in hearing aids. Selected for reliability and compact size. Passive sensing avoids additional power overhead but requires analog conditioning (handled by MCP6232).
MCP6232 Op-Amp (x2)	Low-noise amplification + filtering of telecoil signal	Low power (~100 µA), rail-to-rail, GBW ~300 kHz	Audio Front-End	FUN-03, PER-02, 1.5, 1.10	Gain + frequency response test	Ultra-low power op-amp suitable for battery operation. Sufficient bandwidth for speech signals while minimising current draw. Chosen over high-speed amplifiers due to power constraints. Enables analogue filtering before ADC to improve SNR.
PCM1808 ADC Module	Converts analogue audio (mic/telecoil) → digital (I²S)	24-bit ADC, up to 96 kHz, SNR ~99 dB	Audio Interface	FUN-01, FUN-07, PER-01, 1.6, 1.11	I²S data validation + audio quality test	High dynamic range ensures speech clarity and future-proofing. Native I²S output simplifies STM32 integration. Chosen over MCU-integrated ADCs due to significantly better audio performance (noise, resolution).

Subsystem(s)	User controller subsystem, System monitoring subsystem
Group Member	Anna

ID	Requirement Statement	Verification	Justification / Design Decision	Input	Output	Linked System Req.
			3 User Control Subsystem			
3.1	The user interface subsystem shall detect a rotary encoder with a minimum resolution of ≥24 ticks per revolution.	Rotate the encoder through a full revolution and log the detected step count via the STM32 IDE serial monitor.	To provide sufficiently fine volume control while maintaining mechanical robustness and usability for users.	Rotary Encoder quadrature signals (A and B).	Increment/ decrement the volume control value.	INT-01
3.2	The user interface subsystem shall correlate clockwise rotary encoder rotation to an increase in output volume, and anti-clockwise rotation to a decrease in output volume during normal operation mode.	Functional test by rotating the encoder and measuring the corresponding change in output sound pressure level.	Align with conventional user expectations.	Rotary Encoder quadrature signals.	Volume increase/decrease command.	INT-01, SAF-02
3.3	The user interface subsystem shall ignore rotary encoder rotational input when system is in the Power OFF state.	Attempt encoder rotation while powered OFF and power it ON to see volume response.	Prevent unintended volume change during power OFF state and protect user from unexpected surge in volume.	Rotary Encoder quadrature signal.	No output, since input is ignored.	INT-01
			4 System Monitoring Subsystem			

4.1	The capacitive touch sensor shall detect user proximity within a proven distance if the thirteenth electrode is used.	Verify proximity detection sensing using non-conductive calibrated spacers.	Enables automatic wave and sleep functionality based on wear detection.	Capacitive sensor proximity signal.	Proximity detected or not detected flag.	INT-02
4.2	The system shall transition from sleep mode to normal operation within $\leq 500ms$ when proximity is detected.	Measure elapse time between proximity detection and audio output activation.	Ensure smooth user experience.	Proximity detected flag.	Wake command to system control.	INT-02, POW-02
4.3	The capacitive touch module shall support tap-based input to enable Bluetooth, Telecoil, or a microphone array.	Perform repeated tap gestures and verify Bluetooth mode activation, Telecoil, and ambient audio. Only one mode can work at a time.	Provide intuitive mode switching without additional complex control.	Capacitive touch pad input.	Bluetooth or telecoil or ambient audio from microphone array mode enable command.	INT-03
4.4	The subsystem shall prioritise proximity detection over capacitive touch user input.	Trigger proximity detection simultaneously with the encoder input and observe the priority handling.	Ensure reliable wake-up from sleep behaviour even in the presence of other inputs (such as enabling the different modes)	Proximity signal, touch inputs.	State transition precedence	INT-01
4.5	The system shall enter low power sleep mode when no proximity is detected for a continuous duration of $\geq 30s$.	Leave the device idle and measure the current drawn and state transition timing.	It extends battery life.	Proximity signal, touch inputs.	Reduced current draw.	INT-02, POW-02

Justification of Chosen Development Board

Bourns PEC11R-4215F-S0024 incremental encoder

The Bourns PEC11R-4215F-S0024 incremental encoder was chosen for the volume control because it aligns with subsystem requirements 3.1, 3.2, and 3.3, as well as system requirements INT-01 and SAF-02. It is a through-hole, horizontal PC-pin encoder, making it suitable for PCB mounting with the shaft facing outward for easy user access. Its detent torque of 30-90 gf·cm provides enough resistance to prevent accidental volume changes while still allowing smooth, intuitive control by feel, which is important for hearing aid users who cannot rely on visual feedback.

The selected 24 pulse per revolution configuration satisfies the minimum requirement of 24 ticks per revolution, allowing each detent to correspond directly to one volume step. The encoder's 2-bit quadrature output supports both clockwise and anticlockwise rotation, enabling predictable volume increase and decrease. This makes the control simple, safe, and easy to use. Because the output is digital, it can be read directly by the STM32 GPIO pins without requiring an ADC, reducing both complexity and power consumption. The encoder also has a minimum rotational life of 30,000 cycles, which is adequate for long-term wearable use.

Other options were considered but were less suitable. Potentiometers were rejected because they do not provide discrete steps, require ADC input, and have end stops, which limits continuous adjustment. Capacitive touch sliders, such as the TIDM-02004 approach, offer high-resolution input, but they depend on continuous sensing and gesture interpretation, which increases complexity and makes fine control less deterministic. They also provide less tactile feedback and can be affected by moisture, sweat, and unintended contact. In comparison, the rotary encoder gives discrete, predictable steps with mechanical feedback, making it the better choice for this application.

Adafruit MPR121 12-Key Capacitive Touch Sensor Module

The MPR121 module was selected for the system monitoring subsystem as it aligns with requirements 4.1 to 4.5. The device provides 12 capacitive touch inputs along with a 13th simulated proximity electrode, allowing for both discrete touch input and near-proximity detection within the same module. This is important because subsystem requirement 4.1 requires reliable proximity detection for automatic wake/sleep feature, and the modules 13th electrode is specifically designed to increase sensing area and detection distance.

Power efficiency is achieved through 1.71 - 3.6 V operation (requires 3.3 V regulation from 3.7 V Lithium Polymer battery) with 29 µA run current at Electrode Sample Interval (ESI) of 16 ms and 3 µA stop mode [MPR121 Datasheet]. The ESI=16 ms setting delivers fast response (~60 ms scan cycles) for immediate wake detection (requirement 4.2).

The I²C Fast Mode interface (400 kHz) and Interrupt Request (IRQ) pin enable event-driven STM32 operation without continuous polling. The 400 kHz speed reads all 13 channels in ~0.3 ms versus 1.2 ms at 100 kHz, minimizing active time per event to maximize battery life during proximity checks and mode taps [24].

Capacitance measurement uses 10-bit analog-to-digital converter (ADC) with autoconfiguration of Charge Discharge Current (CDC, 0-63 µA) and Charge Discharge Time (CDT, 0.5-32 µs) per electrode. These optimize V_charge = (I × T)/C for maximum sensitivity across varying electrode capacitances, eliminating manual calibration [AN3889].

Noise immunity includes touch/release thresholds (0-255 counts, typical 4-16), debounce (0-7 consecutive samples), and baseline filtering (Maximum Half Delta/Noise Half Delta 1-63 counts, Noise Count Limit 0-255) for environmental drift compensation from humidity and temperature. [MPR121 Datasheet][25][26]. Alternative considered: The TTP224 capacitive touch module provides only 4 touch pads with no proximity capability [27]. It requires 4 General Purpose Input/Output (GPIO) pins versus MPR121's 2-wire I²C connection, consumes more microcontroller resources, and needs physical pad changes for sensitivity adjustments. MPR121 enables software tuning for humidity, temperature, and electrode variations [27].

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Acknowledgements

Policy on Generative AI

As this assessment task involves some planning or creative processes, you are permitted to use software to generate initial ideas. However, you must develop or edit those ideas to such a significant extent that what is submitted is your own work, i.e. only occasional AI generated words or phrases may form part of your final submission. It is a good idea to keep copies of the initial prompts to show your lecturer if there is any uncertainty about the originality of your work.

If the outputs of generative AI such as ChatGPT form a part of your submission, it will be regarded as serious academic misconduct and subject to the standard penalties, which may include 00FL, suspension and exclusion.

Appendix: Updated Risk Analysis Incorporating Feedback

Include all of these sections from Prototype Design 1 (apart from the system block diagram), updated to account for lecturer feedback. These sections are not graded directly but may be referred to when grading other sections.

ZEIT4230: Prototype Design 1

Group Name	Quiet Riot
Name of Product or System	Digital Hearing Aid
Brief Description of System's Features and Functions (4-5 lines)	
A wearable hearing-assistive system that captures and digitally processes sound from multiple sources including an I2S MEMS microphone for regular conversations, Bluetooth connectivity to mobile devices, and a Telecoil for compatibility with hearing-loop systems. The STM32 processes incoming digital audio and applies filtering and amplification for certain frequencies prior to output. User interaction is provided via a rotary encoder for volume and mute control and a capacitive touch interface for fine-tune adjustments. The output is delivered via I2S to either a conventional speaker or a bone conduction transducer to allow for flexible listening modes in different environments.	

It is expected that your prototype system will consist of **a single STM32 Nucleo board** for the whole group, and at least **one breakout board/development board per group member**. The development boards should all be available either from Digikey's Australian Website, or from Core Electronics. Other suppliers can only be used if you have obtained permission in advance from lecturers.

For this prototype system, you may use the lab power supplies, and you are not required to develop your own custom power supply solution.

STM32 Nucleo board part number	NUCLEO-L476RG

Bill of Materials and Budget

For your prototype system, each group will be allocated a budget of AU$100 x num group members. You can divide this budget amongst your group however you want.

Team Member	Development Board Description	Supplier Name	Supplier Part Number	Individual cost (AUD, excluding GST)	Quantity
Mariya	Adafruit PDM MEMS Microphone Breakout	Core Electronics	ADA3492	$8.95	2
Anna	Bourns rotary encoder	Digikey	PEC11R-4215F-S0024-ND	$3.27	1
	Adafruit capacitive touch module	Digikey	1528-1038-ND	$7.95	1
Ryan	Bluetooth development module	Core Electronics	BT401v3	$26.00	1
	Knowles Telecoil*	Digikey	5100-253447	$9.20	1

	Low power LNA + Filter*	Digikey	MCP6232	$0.56	2
	PCM1808 ADC*	Amazon	PCM1808	$12.64	1
Naomi	Bone conductor speaker	Digikey	1528-1948-ND	$13.22	2
	MAX98357 I2S Amplifier Module	Core Electronics	DFR0954	$9.23	2
	Thin Speaker - 4 Ohm, 2.5W, 28mm	Core Electronics	COM-21311	$3.41	2

This component/module is an extension of the subsystem

Total Budget (AUD, ex-GST)	$193.90

Notes:

- If one group member's development board is very simple (e.g. it doesn't contain any ICs which must be configured via the microcontroller), then it would be appropriate for that group member to be responsible for multiple development boards.
- If you require Nucleo boards different from those supplied, then you will need to factor in their purchase cost in your budget above.

Microcontroller Resource Usage

You must ensure that all of your group's chosen development boards can be connected to your microcontroller at the same time. It is suggested that you use the STM32CubeIDE software to graphically configure the microcontroller's pins and peripherals to ensure that there is no conflict.

GPIO Pin Table

Development Board	Microcontroller Peripheral	GPIO Pins
Adafruit MPR121 12-Key Capacitive Touch Sensor Breakout	I2C1_SCL I2C1_SDA	PB6 PB7
PEC11R-4215F-S0024 Rotary Encoder	TIM2_CH1 (GPIO) TIM2_CH2 (GPIO) GPIO_Input	PA0 PA1 PA12
Adafruit PDM Microphone Breakout (x4 microphones)	DFSDM1_CKOUT (shared clock) DFSDM1_DATIN3 DFSDM_DATIN4	PC2 PC7 PC0
BT401 Bluetooth Module	SAI1_FS_A SAI1_SCK_A SAI1_SD_A	PB9 PB10 PC3
Knowles Telecoil and analog front-end+ADC	SAI2_FS_A SAI2_SCK_A SAI2_SD_A	PB12 PB13 PB15
MAX98357 I2S Amplifier Module	SAI2_FS_B SAI2_SCK_B SAI2_SD_B	PA15 PC10 PC12

For any microcontroller peripherals and GPIO pins that are used by multiple boards, explain how they are shared:

Shared Peripheral/Pins	Explanation
Nil	N/A

Software Risk Analysis (Review)

Confirm the following points

- Will each team member have a substantial piece of code that they can present as their own contribution?
- For every piece of code that you will need to produce, can you find an existing example that you can adapt?

Team Member	Embedded Code to Develop	Existing Code and Steps Needed to Adapt
Mariya	- Sound capture, amplification of quiet sounds, limiting boost of loud sounds. - Noise reduction (remove background hum etc.) - Voice enhancement (detect speech frequencies). - *Use beamforming: Directional listening (tell microphone array which direction to listen to), reducing background noise, improve SNR.	https://youtu.be/QgecXVmNyiI?si=voLpV_9NCaublXQ6 (All initialisation and configuration parameters for PDM conversion) Subsequent videos part of the series of explain connecting of PDM microphone to filtering, decimation etc. Application note AN3998 contain code snippets for PDM filter https://www.st.com/resource/en/application_note/an3998-pdm-audio-software-decoding-on-stm32-microcontrollers-stmicroelectronics.pdf
Anna	Configure the rotary encoder to provide user control over key functions. Rotational	https://deepbluembedded.com/stm32-timer-encoder-mode-stm32-rotary-encoder-interfacing/ This link provides Rotary Encoder Interfacing with STM32. This source uses a Nucleo32-L432KC board, the peripherals and pins used need to be changed in the code to be transferred to our L476RG board. adafruit-mpr121-12-key-capacitive-touch-sensor-breakout-tutorial.pdf (page 24 lists a micro python

	input should be used to adjust volume levels, while the integrated push-button switch should be used to power the device ON and OFF. Configure the capacitive touch sensor module to enable proximity-based wake functionality. The device should remain in a low-power sleep mode when no proximity (e.g., contact with or near the user's skin) is detected, and automatically wake when proximity is sensed. The touch interface	code to test the MPR121 capacitive touch sensor. To adapt the code, we should generate a project with HAL drivers. https://embeddedwiki.com/articles/template1-with-mpr121/touchclamp-click-uni-ds-v8-necto-stm32f107vc (Also combines MPR121 with STM32F107VC. Need to initialise I2C using STM32CubeMX-generated HAL code. Call TouchClamp driver functions using the HAL I2C handle, since this code uses TOUCHLAMP_MAP_MIKROBUS (cfg, MIKROBUS_1) to map I2C, INT and RST pins to mickroBUS socket.)

	should also be used to control features such as enabling or disabling Bluetooth connectivity and telecoil functionality.	
Naomi	Configuration of Class D amplifier and custom frequency-specific amplification. Bone conduction capability*.	The class D amplifier has been used previously on STM32s (reference: STM32F446RE MAX98357A I2S Sound). Code and circuit setups for the STM32 with the class D audio amplifier are here: https://blog.embeddedexpert.io/?p=2553 The bone conductor has the same characteristics as the speakers and so, is readily able to be swapped on the output* (bone conductor datasheet)
Ryan	Configuration of Bluetooth module and audio streaming via I2S. Telecoil audio streaming into the SM32*.	The BT401 v3 is well documented here BT401 Dual Mode Bluetooth Audio Data Transmission BLE Module I2S Audio Output Description V1.6 - Zhihu. The Telecoil is a simple receiver that outputs an analog signal which can then be amplified, filtered, digitised, and then processed (reference: Factsheet by the American Academy of Audiology)*.

This functionality is an extension of the subsystem

- Note any higher-level processing tasks that you think will be needed, and give example where others have previously implemented such tasks on a similar microcontroller

Task	Previous Example

Quantitative Measurements and Instruments

For each member of the group, outline:

- What quantitative performance aspects they could measure during their individual demonstration?
- What instruments, tools or additional hardware would be required, and can you get access to them? (consider both those needed for quantitative measurements, and those needed just to get the system working)

Team Member	Quantitative Measurements	Tools or Instruments Required
Mariya	Single PDM microphone:	• Acoustic calibrator

	SNR Sensitivity, SNR, range between noise floor and maximum sound pressure level. Latency between sound arriving at microphone and when processed signal is available. Beam forming direction of arrival (estimating where sound comes from)	• Signal generator • Sound level meter • Oscilloscope • Sound source • Angle measurement set-up • Matlab
Anna	**For the rotary encoder:** Angular position (angle vs. Time or position vs. Step index) - how far the encoder has rotated from a reference point. Direction of rotation using quadrature signals: if signal A leads B the clockwise, vice versa then it is anticlockwise. Counts per revolution of encoder to quantify mechanical consistency. **For the capacitive touch sensing module:** Touch strength (10 bit values ranging from 0-1023, baseline register 0x1E – 0x2A. Compute Delta = baseline – filtered where filtred is	• Oscilloscope/ Logic Analyser • Multimeter • Matlab • Serial monitor to read digital output values of the capacitive touch module

	measured signal a baslne is no touch referecne from 0x1E-0x2A. Touch sensitivity (per electrode): measure Delta = baseline – filtered (counts) for no-touch vs touch. SNR using the delta values (Noise: standard deviation of delat when there is no touch and Signal: mean when delta there is a touch). Jitter/stability: log delta over time with no touch. Compute standard deviation and peak to peak vaiation SNR, Hysteresis, False Trigger Rate, Response Time Proximity Range/ Sensitivity for proximity detection.	
Naomi	Total Harmonic Distortion+Noise, Gain, bone conductor performance	• Oscilloscope with FFT capability • Sound level meter • Anechoic chamber • Matlab
Ryan	Bluetooth range testing, latency, performance analysis, Telecoil strength testing*	• Ruler • RF Chamber • Analog Discovery • Auditorium with Telecoil capability* • Matlab

Supply-Chain Risk Analysis

In Semester 2 you will be turning this prototype system into a single printed circuit board. List all the key parts on the board that you will need to source for your PCB, and make sure that there are plenty of them in stock at Digikey, or that you have some alternatives

- Most resistors and capacitors can easily be replaced with different parts, so there is no need to list them unless they are very exotic values, high tolerance etc.iuy

Part Description	Manufacturer Name and Part Number	Digikey Part Number	Digikey Current Stock Level	Alternative Part or Supplier Details
MAX98357 IC AMP CLASS D MONO 3.2W 16TQFN	Analog Devices Inc./Maxim Integrated, MAX98357AETE+T	MAX98357AETE+TCT-ND	31,221	From Mouser, 700-MAX98357AETE+T is the same component.
MP34DT05TR-A (PDM MEMS chip)	STMicroelectronics	497-17809-1-ND	104,590	Mouser Electronics
ADC/AUDIO 24BIT 96K 14TSSOP	Texas Instruments, PCM1808PW	296-19773-ND	607	Mouser alternative or PCM1808PWR.
Amplifier IC 1-Channel (Mono) Class AB 14-TDFN (3x3)	Analog Devices, MAX9814ETD+T	MAX9814ETD+TCT-ND	2624	Mouser alternative
Audio & BLE/SPP Pass-through Module -	DFRobot, BT401v3	DFR0781	Core electronics (multiple available)	Amazon alternative

Bluetooth 5.0				
IC OPAMP GP 2 CIRCUIT 8DIP	Microchip, MCP6232	MCP6232-E/P	869	Digikey alternative
Capacitive Touch Module 12 QFN-20(3x3) Sensor, Capacitive Touch RoHS	NXP MPR121QR2	C91322 (only available on LCSC) Electronics	3986 (In-stock only on LCSC Electronic, Digi key discontinued this part)	None
Rotary Encoder Mechanical 24 Quadrature (Incremental) Vertical	Bourns Inc. PEC11R-4215F-S0024	PEC11R-4215F-S0024-ND	10, 903	From Mouser, Mouser No: 652-PEC11R-4215F-S24, which is the same as the part from Digikey

Manufacturability

- Make sure that it will be possible to hand solder and modify the chosen parts if needed - avoid leadless packages such as BGA or QFN
- Ensure that the required track width and spacing of the PCB will be compatible with standard PCB manufacturing processes, and will not require something more expensive and exotic
- As a point of reference, the standard PCBWay process has a minimum track width and track spacing of 0.16mm, so anything above those dimensions should be straightforward

Part	**Package**	**Pin Pitch and width**	**Smallest track width and spacing needed**
NXP MPR121QR2 (Capacitive sensor)	QFN-20(3x3)	0.4mm/0.2mm	0.05mm/0.075mm
PEC11R-4215F-S0024 (rotary	Through hole	2.5mm/0.8mm	0.25mm/ 0.25mm
MP34DT05TR-A (PDM microphone)	HCLGA 4LD (3 x 4 x 1 mm)	1.6mm 0.8mm	0.2mm
MAX9814ETD+T	14-TDFN	0.4mm/0.2mm	0.2mm/0.2mm

- Make a note of any high speed digital signals (typically routed as differential pairs) or high frequency RF signals on the development board that would need to be included on your own PCB, and confirm that they look feasible to implement
- Simple RF connections are fine, but laying out your own PCB trace antenna may be too risky, and would be very difficult to debug if it goes wrong

Board Description	High speed/high frequency tracks to route
BT401 Bluetooth Receiver	Manufacturer's guidance on placement. In particular, minimum 15mm clearance from ground planes under the antenna and test point.

Additional Checks

Is your system battery powered? If so, outline how you have chosen low-power parts to suit a battery-powered application.

Y/N	Details
Y	• The system is designed as a fully battery-powered wearable device, requiring strict power optimisation across all subsystems to maximise operational lifetime and minimise thermal output. • The audio amplification stage utilises a low-power Class-D amplifier topology, selected for its high efficiency (typically >85%) compared to Class-AB alternatives, thereby significantly reducing power loss as heat while maintaining adequate audio output performance. • Audio output components (amplifier and speaker) were selected to operate within a constrained power envelope (~2.5 W peak), ensuring sufficient loudness while avoiding unnecessary energy consumption typical of over-specified drivers. • The processing unit (STM32 microcontroller) was chosen for its ultra-low-power modes, enabling dynamic power scaling through sleep and standby states during idle or low-activity periods. • MEMS PDM microphones were selected due to their inherently low current consumption and direct digital output, eliminating the need for power-hungry analogue front-end circuitry (e.g., external ADCs and amplifiers). • Bluetooth functionality is implemented using Bluetooth Low Energy (BLE), which is specifically optimised for low-duty-cycle communication and significantly reduces transmission power compared to classic Bluetooth protocols. • The system incorporates intelligent power management strategies, including selective subsystem shutdown (e.g., disabling Bluetooth or telecoil modes when not in use), reducing average current draw. • Energy storage is provided by $LiFePO_4$ cells, selected for their stable discharge characteristics, high cycle life, and improved safety compared to other lithium chemistries, making them well-suited for continuous wearable operation.

Does any of your development boards need a non-standard PCB material (i.e. other than FR4)?

- Note that this might cause problems when it comes to integrating all development boards onto a single PCB made form a single material

Y/N	Details
N	

Do any of your development boards dissipate significant power, and require heatsinking?

Y/N	Details
N	

What is the maximum output voltage in your system, and where does it occur? (Note that this must be less than 40V for safety reasons)

5V output supply to audio amplifier, output of amplifier up to 5V peak.

- Do any of your development boards contain any high power light sources (including infra-red or ultra-violet), and if so how do you avoid potential eye hazards?
- Note that the use of lasers is strictly prohibited

Y/N	Details
N	

Development Board Schematics

- All chosen development boards should have openly available schematics that you can adapt into your own circuit board in Semester 2
- Attach copies of these schematics to your submission, so that lecturers can evaluate their complexity and advise if you are taking on too much risk
- If you do not need the full functionality of every development board, and would only incorporate part of it into your own circuit, then note this below
- Confirm that your chosen development boards don't contain another microcontroller that prevents your STM32 microcontroller from directly accessing the parts you need

Student	Development boards	Includes Microcontroller?	Note any parts of the board you plan not to use
Anna	Bourns Rotary Encoder Adafruit Capacitive Touch Module	N	
Mariya	Adafruit PDM Microphone Breakout Board	N	

Naomi	MAX98357 I2S Amplifier Module	N	
Ryan	BT401 Bluetooth module	N	All auxiliary peripherals (play, volume, skip, indicators). Intention to bypass the I2S audio jack and solder directly to the I2S pins.